Unit 2 FRQ Set 2

Forces & Translational Dynamics · 4 free-response questions in real AP exam order

40 points Time: **95 min** (the AP exam's FRQ section) MR 10 · TBR 12 · LAB 10 · QQT 8

- Read all four questions first (2 min). Start with the one you're most sure of.
- **Derive** = start from a law in symbols. **Calculate** = equation, numbers in, units out. **Justify** = claim → law → link to THIS situation.
- Never leave a part blank. Points are awarded line by line.
- When the timer ends, stop. Grade yourself with the coaching guide's rubric, and be strict.

Reference (same ideas as the AP equation sheet)

$\Sigma F = ma$ · $F_g = mg$

$f_s \le \mu_s F_N$ · $f_k = \mu_k F_N$

$x_{cm} = \Sigma m_i x_i / \Sigma m_i$

$F_g = Gm_1m_2/r^2$ · $g = GM/r^2$

$a_c = v^2/r$ · $v = 2\pi r/T$

$F_s = -k\Delta x$

$v = v_0 + at$

$\Delta x = v_0 t + \frac{1}{2}at^2$

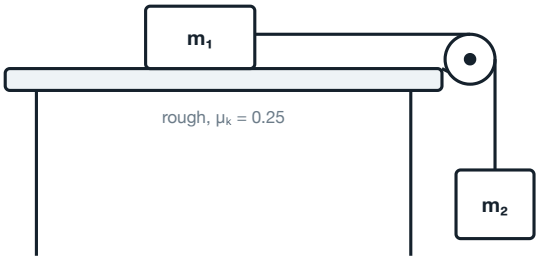
$v^2 = v_0^2 + 2a\Delta x$

$g = 9.8\ m/s^2$ · $G = 6.67 \times 10^{-11}\ N \cdot m^2/kg^2$

Question 1: Mathematical Routines (MR)

10 points · suggested 20–25 min

Block 1 (mass m_1 = 4.0 kg) slides on a rough horizontal table (μ_k = 0.25). It is tied by a light string over an ideal pulley to Block 2 (mass m_2 = 2.0 kg), which hangs. Block 1 moves toward the pulley.



A. On the dot below, which represents Block 1, draw and label the forces exerted on Block 1 as it slides. (2 pts)

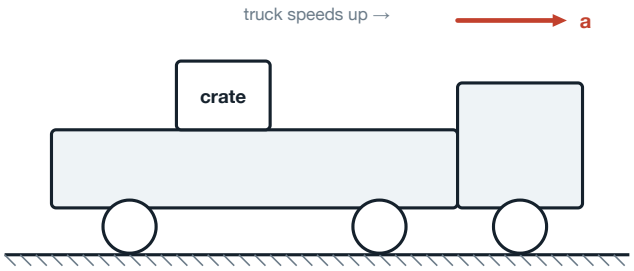
B. Derive an expression for the acceleration of the blocks in terms of m_1, m_2, μ_k and g. Begin with a fundamental principle. (3 pts)

C. Calculate (i) the acceleration and (ii) the tension in the string. (2 pts)

D. Block 1 is replaced by an 8.0 kg block of the same material. The system is given a quick push so that the blocks are already moving, and then released. Describe the motion of the blocks after release. Justify your answer. `(3 pts)`

Question 2: Translation Between Representations (TBR) 12 points · suggested 25–30 min

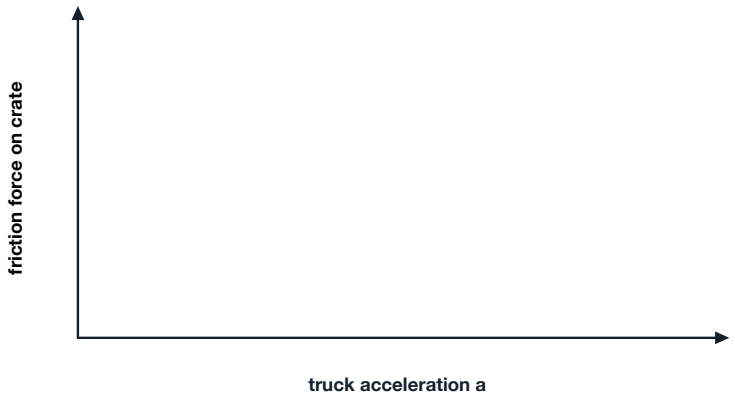
A crate of mass m sits on the flat bed of a truck. The coefficient of static friction between the crate and the bed is μ_s, and the coefficient of kinetic friction is μ_k ($\mu_k < \mu_s$). The truck starts from rest and speeds up with a constant acceleration a toward the right. The crate does not slide.



A. On the dot below, which represents the crate, draw and label the forces exerted on the crate while the truck speeds up. (3 pts)

B. Derive an expression for the maximum acceleration a_{max} the truck can have without the crate sliding. Express your answer in terms of μ_s, m and physical constants, as appropriate. (3 pts)

C. On the axes below, sketch the magnitude of the friction force on the crate as a function of the truck's acceleration a, from a = 0 to a value greater than a_{max}. Label any important values on the axes. `(4 pts)`

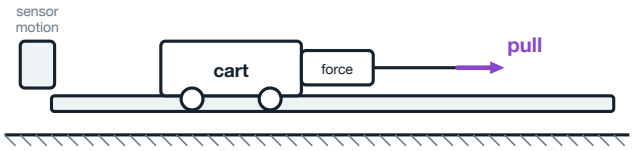


D. The crate is replaced by a crate of mass 2m made of the same material. Does a_{max} increase, decrease, or stay the same? Justify your answer in a way that is consistent with part B. `(2 pts)`

Question 3: Experimental Design and Analysis (LAB)

10 points · suggested 25–30 min

Students want to determine the inertial mass of a cart *without* using a balance. They have a low-friction track, the cart, a force sensor that can pull the cart horizontally, and a motion sensor that records the cart's velocity as a function of time.

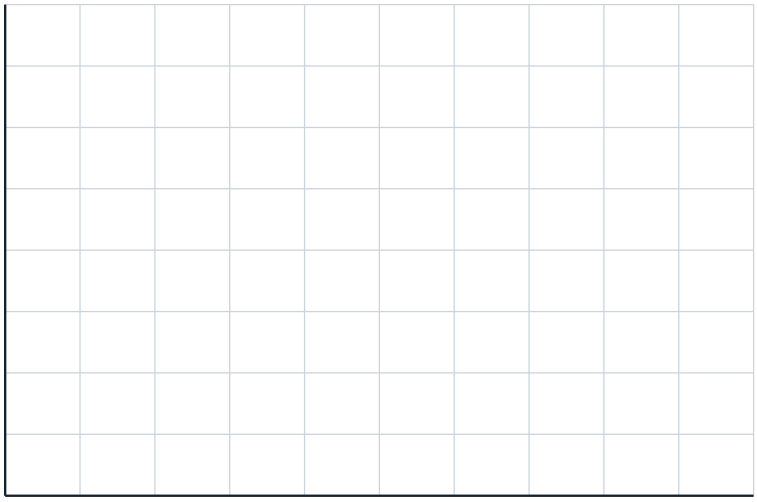


A. Describe a procedure the students could use to collect data to determine the cart's mass using a graph. Include what is varied, what is measured, and how each measurement is made. (2 pts)

B. The data are shown below. Indicate which quantities the students should graph to produce a straight line, and explain how the mass can be found from that graph. (2 pts)

Applied force F (N)	Acceleration a (m/s²)
0.50	0.42
1.00	0.79
1.50	1.27
2.00	1.63
2.50	2.08

C. On the grid below, plot the data, label the axes with units, use an appropriate scale, and draw a best-fit line. **(4 pts)**

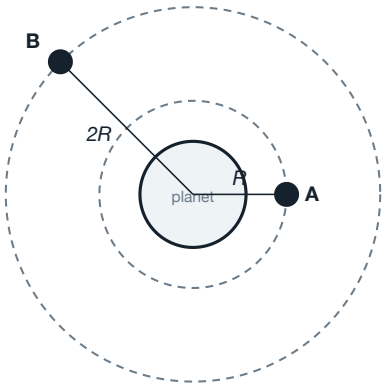


D. Use your best-fit line to calculate the mass of the cart. **(2 pts)**

Question 4: Qualitative/Quantitative Translation (QQT)

8 points · suggested 15–20 min

Satellite A orbits a planet of mass M in a circle of radius R. Satellite B, which has the same mass, orbits the same planet in a circle of radius 2R.



A. WITHOUT using equations, explain whether Satellite B's orbital speed is greater than, less than, or equal to Satellite A's. (3 pts)

B. Derive an expression for the orbital speed of a satellite in a circular orbit of radius r around the planet, in terms of G, M and r. (3 pts)

C. Explain how your expression in part B supports your reasoning in part A, and find the ratio v_B/v_A. (2 pts)